Sensorless Control of Brushless DC Motor with Delta Connection Windings

Jianqi Qiu, Cenwei Shi
College of Electrical Engineering, Zhejiang University, Hangzhou 310027, China
E-mail: motor@zju.edu.cn

Abstract —This paper presents a phase-locked loop (PLL) sensoless control method based on the integration of the resultant back EMF for delta connection brushless dc motors. The mathematical model is established, from which the resultant back EMF is derived. The simulation and the implemental results confirm the feasibility of this method.

Index Terms —Brushless dc motor, delta-connected winding, sensorless control, phase-locked loop.

I. INTRODUCTION

Permanent magnet brushless dc (BLDC) motors are extensively used for industry application due to the high power density, high efficiency and excellent controllability. Y-connected three phase winding is widely used in a BLDC motor. However, delta-connected winding is hardly used because of the circulation current caused by triple harmonics of the back EMF. In order to eliminate the circulation current, a proper design needs to be made based on the number of poles, slots, pole arc and the choice of the back EMF wave shape. From the viewpoint of mass production, the delta-connected three phase winding can be wound continuously using automatic winding machine. Therefore, the cost of a BLDC motor can be reduced by improving productivity [1].

Sensorless control for brushless dc motor drive has been drawing much interest in recent years. The most popular schemes on rotor position sensorless control is based on the back EMF sensing technique, which has been well studied in prior works [2-6]. Generally, the back EMF can be utilized in three ways: 1) detecting the zero crossings of the back EMF, 2) integrating the back EMF from its zero crossing and comparing the integration result with a reference voltage, 3) integrating the back EMF and using a phase-locked loop (PLL) to keep the integration result being zero. However, these methods usually focus on the Y connection motors with six-step commutation sequence, that is, only two phase conduct current at any times, leaving the third phase floating. But for delta connection motors, three phase winding always conduct current at any step of six-step commutation, the back EMF of the floating phase will not be detected.

This paper proposes a phase-locked loop (PLL) sensorless control method based on the integration of the resultant back EMF for delta-connected BLDC motors. The mathematical model of delta-connected BLDC motor with six-step commutation is established, and the phase relationship between the terminal voltage, phase current and line current are analyzed. The simulation and the experimental result confirm the validation of the proposed method.

II. MATHEMATICAL MODEL OF DELTA CONNECTED BLDCM

A. Y Connected BLDC Motor

In most case, a brushless dc motor with Y connected winding is driven by a three phase inverter with six-step commutation. Each step, only two power switches are conducted simultaneously. The commutation sequence is determined by the rotor position. Fig.1 shows the equivalent circuit of a Y connected brushless dc drives.

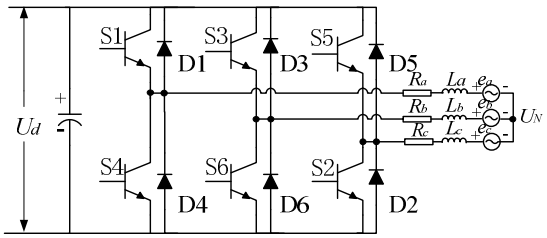


Fig.1 The equivalent circuit of Y connected BLDCM

Assuming at a particular step A+B-, phase A and B are conducting, phase C is floating. The following equations can be obtained from the equivalent circuit.

$$\begin{aligned} U_{dc} &= R_a i_a + L_a \frac{di_a}{dt} + e_a - R_b i_b - L_b \frac{di_b}{dt} - e_b \\ i_a &= -i_b \\ U_N &= -R_b i_b - L_b \frac{di_b}{dt} - e_b \end{aligned} \tag{1}$$

where R_a, R_b, R_c are the phase resistances, L_a, L_b, L_c are the equivalent phase inductance, i_a, i_b, i_c are the phase currents, e_a, e_b, e_c are the phase back-EMF, U_{dc} is the DC link voltage, U_C is the terminal voltage of phase C, U_N is the neutral point voltage of three phase winding. The following equations can be further derived from (1):

$$U_C = e_c + U_N = \frac{1}{2}U_{dc} - \frac{1}{2}(e_a + e_b) + e_c \tag{2}$$

In generally case, considering a symmetrical three phase windings, if only the fundamental component of the back EMF is considered, we can get:

$$e_a + e_b + e_c = 0 \tag{3}$$

Thus, (2) can be rewritten as

$$U_C = \frac{1}{2}U_{dc} + \frac{3}{2}e_c \tag{4}$$

Clearly, U_C contains the back EMF information of phase C, the zero-crossing of phase C back EMF can be obtained from the intersection of the terminal voltage and half of DC link

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voltage. Therefore, the terminal voltage of the floating phase is suitable for sensorless control of a brushless dc motor with Y-connected winding.

B. Delta Connected BLDC Motor

Fig.2 shows the Equivalent circuit of a delta connected brushless dc motor. Assuming the A+B- step, the following equations can be obtained

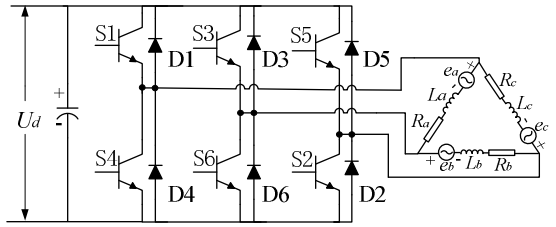


Fig.2 The equivalent circuit of delta connected BLDCM

$$U_{dc} = e_a + R_a i_a + L_a \frac{di_a}{dt}$$
$$U_{dc} = R_b i_b + L_b \frac{di_b}{dt} - e_b + R_c i_c + L_c \frac{di_c}{dt} - e_c \quad (5)$$
$$i_b = i_c$$

In balanced three phase windings, if only the fundamental component of the back EMF is considered, we can get:

$$e_a + e_b + e_c = 0 \quad (6)$$

From (5) and (6), the terminal voltage U_C can be expressed by:

$$U_C = \frac{1}{2}U_{dc} + \frac{1}{2}e_a + e_c \quad (7)$$

It can be seen that the terminal voltage U_C contains the back EMF information of both phase A and phase C simultaneously during the step of A+B-. Therefore, the zero crossing of the back EMF can not be extracted from the terminal voltage U_C directly. The most popular method based on the detection the zero crossing of the back EMF can not be used in delta wound BLDC motor.

According to mathematic transformation, the waveform of the resultant back EMF $(e_a/2 + e_c)$ is still sinusoidal, which is symmetrical with positive and negative half-wave. Fig.3 shows the waveforms of the resultant back EMF and the terminal voltage respectively. During A+B- step, the two waveforms are coincident exactly. The integration result of terminal voltage U_C is zero with the interval across from -30 degree to 30 degree around the neutral point of the bus voltage. So that the approach based on integrating the back EMF waveform can be extended to implement sensorless control of the brushless dc motors with delta connected winding.

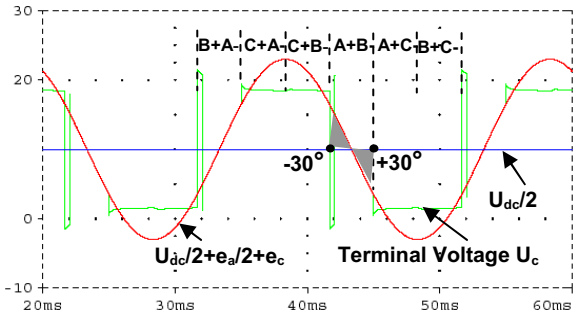


Fig.3 Resultant back EMF and terminal voltage

III. PROPOSED SENSORLESS CONTROL MATHOD

The proposed method integrates the resultant back EMF extracted from the selected terminal voltage and uses a phase-locked loop (PLL) to implement sensorless control of the brushless dc motor. The PLL is formed with the controller, inverter and motor. The system block diagram of the proposed sensorless control strategy is shown in Fig.4.

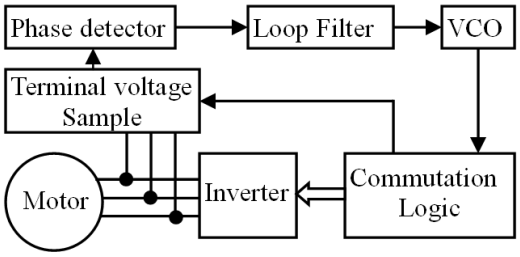


Fig.4 The system block diagram of the proposed method

One of the motor terminal voltages is sampled and integrated over a 60 degree interval. As shown in Fig.3, the integration result is zero if the interval across from -30 degree to 30 degree around the neutral point of the bus voltage, so that the output frequency of the VCO keeps stable. The output frequency of the VCO determines the commutation frequency of the brushless dc motor, hence normal commutation is realized. During the transient state, if the commutation instant is advanced, the integration result will be negative. This will reduce the input voltage of the VCO in the PLL and consequently decrease the motor commutation frequency, which will bring the next commutation closer to the ideal position until the commutation is no longer advanced. In contrast, if the commutation instant is retard, the integration result will be positive, hence the frequency of the VCO will be ramp up until the commutation instant is perfect.

Fig.5 shows the block diagram of the terminal voltage sampling circuits. The neutral point voltage of three phase winding is extracted by a resistor network, and subtracted it from the selected terminal voltages according to six-step commutation logic. The desirable signal can be obtained by changing the polarity of the subtraction result every step. The output signal U_{out} contains the information of the resultant back EMF only, and can be used as input signal of the phase detector.

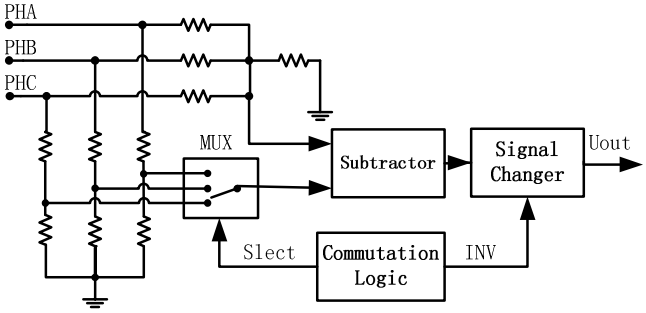


Fig.5 the schematic of the terminal voltage sample circuit

The PLL algorithm can be implemented using CMOS device such as 74HC4046. However, the performance of PLL

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is influenced by the parameters of the peripheral analog component. The phase delay introduced by the hardware integrator will vary with the motor speed. To overcome these drawbacks, a full digital PLL technique is introduced to implement the proposed sensorless control method. A digital multiplier is used as the phase detector, the loop filter is built from an up/down counter, and the VCO is replaced by a digital controlled oscillator (DCO). The entire PLL function is realized by software base on the DSP hardware platform [7].

IV. Simulation Result

The proposed sensorless control method has been verified using Pspice software. Fig.6 illustrates the phase relationship between the commutation signals, the sampling control signals and the polarity control signal respectively. It can be seen that the sampling control signals (S_1, S_2, S_3) and the polarity control signal (INV) can be derived from the six commutation signal (T_1~T_6) through logic operation. Fig.7 shows the waveform of the resultant sampling signal U_{out}. The signal actually contains the pulse due to the freewheel diode conduction, which can be eliminated using digital filter.

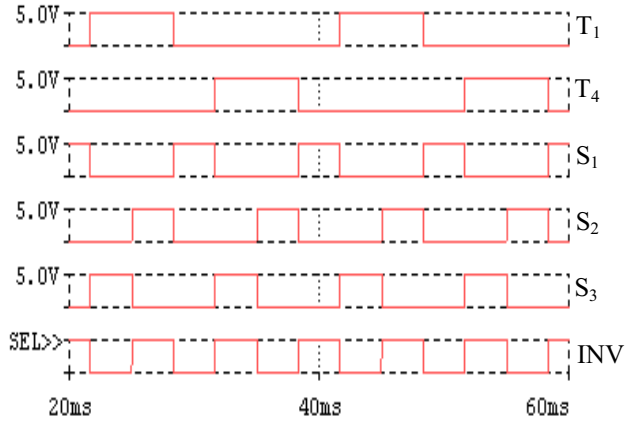


Fig.6 The waveforms of the control signals

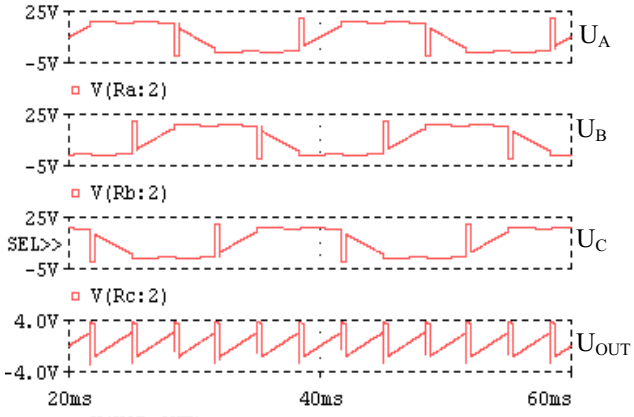
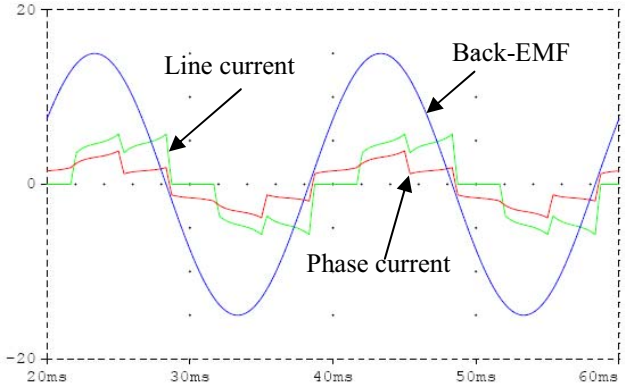
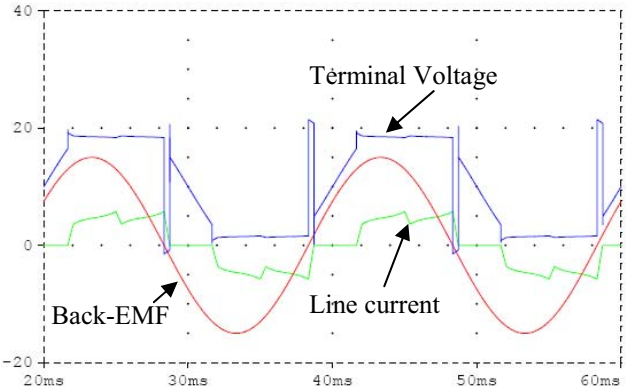


Fig.7 The waveforms of the terminal voltage (U_A,U_B,U_C) and the resultant sampling signal (U_{out})

Fig.8 shows the steady simulation waveforms of a delta-connected BLDC motor driven by the proposed sensorless methods. It can be seen that the phase EMF is in phase with the corresponding phase current to produce the maximum electromagnetic torque. The terminal voltage is in phase with the corresponding line current, and has a phase delay with the phase back EMF, which is different from the Y-connected BLDC motor.



(a)



(b)

Fig.8 Simulation waveforms of the terminal voltage, back-EMF and line current

V. System Implementation and Experimental Results

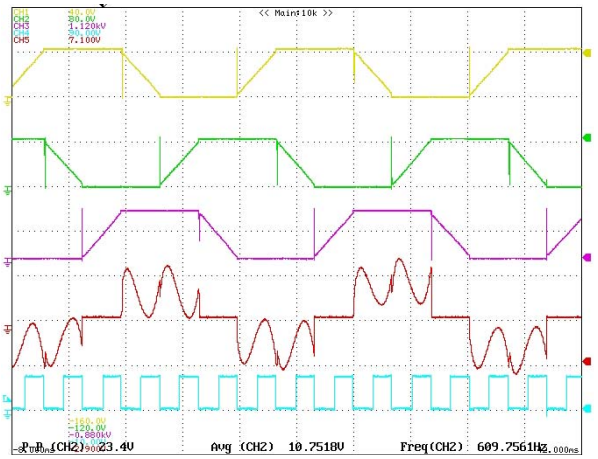
The proposed sensorless method has been implementing using TI TMS320F2812 digital signal processor (DSP). The experimental setup consists of a delta-connected BLDC motor coupled with a PM dc generator. The motor is driven by a three phase power module. The terminal voltages of the three phase winding are sensed and fed to the ADC module of the DSP through a resistor network as shown in Fig.5.

Fig.9 shows the steady state waveforms at sensorless operation. The bottom waveform is the output signal of the VCO, which determined the commutation instants. It can be seen that the line current is in phase with the corresponding terminal voltage under different operation condition. Therefore, the proposed method is always effective whether the motor is running without PWM or with PWM.

Fig.10 shows the starting process of the proposed sensorless method. An open loop start-up procedure is carried out to accelerate the rotor from standstill up to a threshold speed enough for detecting the back EMF. The motor is initially aligned into a fixed position by exciting the three-phase winding, after the forced alignment, the output frequency of VCO is ramped up gradually, which increases the commutation frequency, and therefore the motor speed. At the

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end of ramp-up stage, the open-loop commutation is automatically switched into synchronized commutation mode controlled by PLL. As shown in Fig.10, the switch process is almost seamless, which improves the starting performance of the proposed sensorless control method.



(a)

(b)

Fig.9 Experimental waveforms of the proposed sensorless method: (a) operation without PWM (5ms/div), (b) operation with PWM (10ms/div).

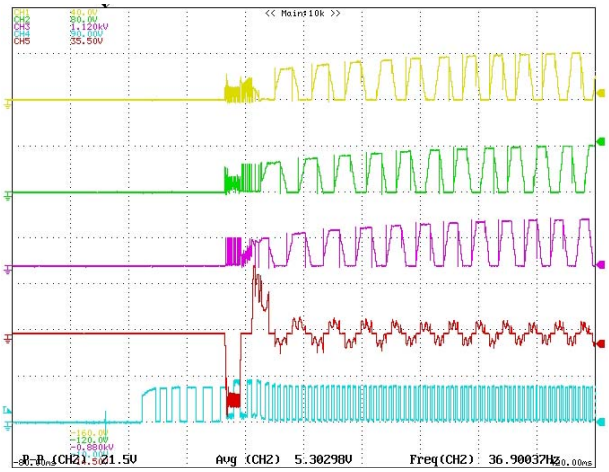


Fig.10 Sensorless operation at starting (from top to bottom: the terminal voltage U_A, U_B, U_C, line current, output signal of VCO)

VI. Conclusion

A PLL sensorless control method has been successfully implementing on a delta-connected BLDC motor. The information of the back EMF containing in the terminal voltage of a delta-connected BLDC motor is analyzed through mathematical model. Due to the lack of unexciting phase winding under six-step commutation mode, the resultant back EMF instead of phase back EMF is extracted from the selected terminal voltage, a digital phase-lock loop technique is introduced to synchronize the commutation instants with the back EMF signals. The validation of the proposed method is confirmed by the simulation and the experimental result.

References

[1] J. R. Hendershot Jr, T. J. E. Miller, Deign of brushless permanent-magnet motors, Oxford New York, 1994.

[2] J. Johnson J.P., Ehsani M., Guzelgunler Y., "Review of sensorless methods for brushless DC," IEEE, IAS1999, Vol.1, pp.143-150, 1999.

[3] P. P. Acarnley, J. F. Waston, "Review of position-sensorless operation of brushless permanent-magnet machines," IEEE Trans. Ind. Electron., vol.53, no.2, pp.352-362, Apr. 2006.

[4] A. B. Plunkett, "Back-EMF sampling circuit for phase locked loop motor control," U.S. Patent 4 928 043, May 22, 1990.

[5] G. J. Su, J. W. McKeever, "Low-cost sensorless control of brushless dc motors with improved speed range," IEEE Trans. Power Electron., vol.19, no.2, pp.296-302, Mar. 2004.

[6] J. X. Shen, S. Iwasaki, "Sensorless control of ultrahigh-speed PM brushless motor using PLL and third harmonic back EMF," IEEE, Trans. Ind. Electron., vol.53, no.2, pp.421-428, Apr. 2006.

[7] Roland E. Best, Phase-locked Loop: Design, Simulation, and Application (5th Edition), McGraw-Hill, 2003.

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